

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Cancer	1			1		
	(b)	(i)	I	all 5 points correct = 2 marks 4 points correct = 1 mark 0/1/2/3 points correct = 0 marks Tolerance +/- ½ small square		2		2	2	
			II	straight line connecting each point <u>including</u> <u>2009</u>		1		1	1	
		(ii)		Any one from • {line/%/it/ results} did not fall {until after 2008/ until 2009} • {line/%/it/ results} stayed the same in 2008 • values in (2007 and) 2008 {were the same/ did not change} • {line/%/it/ results} stayed the same for a year			1	1		
		(iii)	I	Continuing trend from plotted points with straight line using a ruler from 2014 – 2020 + correct answer 13/ from candidates graph		1		1	1	
			II	the (rate of) decline is <u>constant</u> / continues to drop at 1% per year or description of ECF if pattern shown on graph rather than straight line Accept ‘the pattern is repeated’ NOT continues to decrease			1	1		

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		(iv)		One (x1) from: raise price/excise duty/tax (1) education on dangers(1) encourage use of alternatives (1) raise age for buying cigarettes (1) plain packaging (1) Remove from display (1) NOT ban smoking		1		1		
5	(c)			<i>Indicative content</i> • two groups/ smokers and non smokers • measure breathing rate • repeat • find the mean • compare the two groups • two fair testing – • e.g. same age/ fitness/ gender/ length of time smoking/ number smoked per day/ whether they had exercised prior to test 5 – 6 marks : Detailed description of the entire investigation <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>			6	6		6

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				3 – 4 marks : Outline general description of the investigation <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> <i>The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1 – 2 marks : Partial description of investigation <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> <i>The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation</i> 0 marks: No attempt made or no response worthy of credit.						
				Question 5 total	1	5	8	14	4	6